ODOO HACKATHON 2026 | GREEN & RENEWABLE ENERGY TRACK | AUTONOMOUS O&M ENGINE | UAE / DESERT TECH

# DustWatch Autonomous Solar Soiling & Cleaning Optimizer

An intelligent financial-decision engine inside Odoo that converts solar panel dust accumulation into exact Dirham losses, automatically booking cleaning jobs the exact day cleaning becomes cheaper than waiting.

> *"Today owners clean by the calendar. DustWatch lets them clean by the numbers."*

| 12.7% – 27% | AED 8,500 | 725+ MW | 0.0 Liters |
|---|---|---|---|
| UAE OUTPUT DROP TO DUST | ANNUAL SAVINGS PER 500 KWP ROOF | SHAMS DUBAI CONNECTED MARKET | ZERO WASTED WASH WATER |

## 1. EXECUTIVE SUMMARY & CORE VALUE PROPOSITION

Solar panels in arid environments like the UAE quietly lose generation capability every single day due to atmospheric dust and sand deposition. Traditional maintenance relies on arbitrary fixed calendar intervals (e.g., monthly or quarterly), which inevitably burns operating capital through unnecessary washings or forfeits massive revenues through unmitigated soiling.

**DustWatch** solves this dilemma directly inside Odoo by providing an autonomous, closed-loop financial optimization engine:

- **Prices the Energy Deficit:** Compares actual inverter generation against expected clean output under real sunlight hours and converts the kilowatt-hour shortfall into local currency (AED).
- **Optimal Breakeven Trigger:** Maintains a running cumulative loss tally and automatically generates a cleaning work order when accumulated financial losses equal the cost of a single cleaning intervention.
- **Audits Verification & Payback:** Evaluates post-clean generation recovery to prove client ROI and distinguish equipment failure from dust accumulation.

## 2. TEAM STRUCTURE & 4-PERSON DIVISION OF LABOR

The 4-member hackathon pod is split into two specialized, synchronized pairs with clearly defined ownership:

### Pair 1: The Frame (Technical & Odoo) — 2 Developers

**Mission:** Develop the Odoo 19 module, calculation algorithms, and live demo scenario.

- **Member 1 (Math & ORM Lead):** Defines models (`dustwatch.site`, `reading`, `job`), implements Python calculation methods (6.1–6.7), and builds the CSV data loader for 90-day time-series readings.
- **Member 2 (Odoo UI & Workflow Lead):** Constructs color-coded Site Tree view (Green/Amber/Red badges), 2-line performance graph, and Cleaning Kanban Board with Odoo Project/Contact integration.

### Pair 2: Presentation (Canva & Pitch) — 2 Designers

**Mission:** Design an elite 12-slide Canva deck, articulate ROI, and rehearse judge defense.

- **Member 3 (Canva Pitch Lead):** Creates the 12-slide master deck in Canva with desert problem infographics, process loops, visual hierarchy, and crisp typography.
- **Member 4 (Business Model & Q&A Lead):** Validates economic comparisons (AED 8.5k savings/roof), crafts the water conservation angle, writes the live demo script, and leads judge Q&A defense.

### 48-Hour Sprint Synchronization Milestones

**Hour 12:** Frame team locks Odoo models; Canva team locks pitch storyline, slide deck layout, and visual identity.
**Hour 24:** Frame team verifies calculation pipeline & CSV ingestion; Canva team completes draft slides & infographics.
**Hour 36:** Joint rehearsal: Technical demo seamlessly executed on Slide 7. Pitch locked at strictly 5:00 minutes.

## 3. THE PROBLEM: THE DESERT SOLAR DILEMMA

Solar panels only generate electricity when light penetrates the protective glass. In the UAE's hyper-arid desert climate, particulate sand and dust deposition create a steady, compounding degradation in photovoltaic performance:

**12.7% Drop in 5 Months**
UAE field studies documented an average 12.7% output drop when panels were left unwashed over a dry 5-month period.

**27% Peak March Loss**
Abu Dhabi research recorded output losses peaking at 27% during March sandstorms, accumulating at nearly 1% daily.

**725+ MW Connected**
Dubai's Shams Dubai initiative has deployed solar across 8,430+ buildings, representing a vast, underserved maintenance base.

**Failure Mode A: Cleaning Too Rarely**
Compounding dust reduces generation week after week. By month three, forfeited electricity value vastly outstrips the fee for a professional clean.

**Failure Mode B: Cleaning Too Often**
Fixed weekly washings incur inflated contractor invoices and squander thousands of liters of scarce desalinated water washing clean panels.

## 4. THE MATHEMATICAL ENGINE (CALCULATIONS 6.1 – 6.7)

DustWatch replaces arbitrary calendar schedules with a transparent, deterministic mathematical decision framework:

**6.1 Expected Daily Electricity Output (Clean Baseline)**
Theoretical generation of pristine panels under actual site sunlight hours

```
Expected kWh = Capacity (kWp) × Sun Hours × PR (0.80)
```

**6.2 Electricity Lost to Dust (Daily Soiling Deficit)**
Physical shortfall between expected and inverter-measured generation

```
Lost kWh = max( 0, Expected kWh - Actual kWh )
```

**6.3 Daily Financial Loss**
Converts abstract kWh deficits into concrete currency losses for the property owner

```
Money Lost (AED) = Lost kWh × Electricity Tariff (AED/kWh)
```

**6.4 & 6.5 The Optimal Cleaning Trigger (Breakeven Rule)**
Dispatches cleaning work order the exact day accumulated losses hit cleaning cost

```
Trigger Job IF: Cumulative Loss (AED) ≥ Cost of Clean
```

**The Mathematical Breakeven Proof**
Total cost of soiling over time equals: *Total Cost = Cleaning Invoices + Value of Lost Electricity*. Under steady dust deposition, the sum of these opposing cost functions forms a convex curve with a global minimum occurring precisely when accumulated energy loss matches the cost of a single clean. Cleaning prior to this point overpays for labor; cleaning after overpays in forfeited electricity.

**6.6 Recovery & Payback Speed**
Verifies output rebound & calculates payback days

```
Payback Days = Clean Cost / Recovery_Day
```

**6.7 Dynamic Baseline Learning**
Auto-calibrates PR based on 3-day post-clean output

```
PR_new = Avg(Actual / (Cap × Sun))
```

## 5. WORKED EXAMPLE: 500 KWP HOTEL ROOFTOP SIMULATION

Parameters: 500 kWp system, AED 0.40/kWh tariff, AED 1,500 cleaning quote, 5.5 sun hours/day, 0.2% daily dust accumulation:

| Day | Expected Output | Actual Output | Lost to Dust | Daily Loss (AED) | Running Loss (AED) | Odoo Alert Status |
|---|---|---|---|---|---|---|
| Day 1 | 2,200.0 kWh | 2,195.6 kWh | 4.4 kWh (0.2%) | AED 1.76 | AED 2 | Healthy (<50%) |
| Day 10 | 2,200.0 kWh | 2,156.0 kWh | 44.0 kWh (2.0%) | AED 17.60 | AED 97 | Healthy (<50%) |
| Day 20 | 2,200.0 kWh | 2,112.0 kWh | 88.0 kWh (4.0%) | AED 35.20 | AED 370 | Healthy (<50%) |
| Day 30 | 2,200.0 kWh | 2,068.0 kWh | 132.0 kWh (6.0%) | AED 52.80 | AED 818 | Warning (>50%) |
| Day 40 | 2,200.0 kWh | 2,024.0 kWh | 176.0 kWh (8.0%) | AED 70.40 | AED 1,443 | Limit Approaching |
| **Day 41 (Trigger)** | **2,200.0 kWh** | **2,019.6 kWh** | **180.4 kWh (8.2%)** | **AED 72.16** | **AED 1,515** | TRIGGER: Cleaning Job Auto-Booked |
| **Post-Clean (Day 42)** | **2,200.0 kWh** | **2,190.0 kWh** | **10.0 kWh (0.5%)** | **AED 4.00** | **AED 0 (Reset)** | Recovery: AED 68/day (23-Day Payback) |

## 6. ODOO TECHNICAL ARCHITECTURE (THE FRAME POD)

DustWatch maintains an agile footprint by implementing three core models that integrate natively with Odoo:

**1. Site Model (`dustwatch.site`)**

- `name`: Installation identifier
- `partner_id`: Customer (`res.partner`)
- `capacity_kwp`: Solar size (kWp)
- `tariff_aed`: Electricity rate
- `clean_cost_aed`: Cleaning quote
- `pr_factor`: Performance ratio (0.80)
- `running_loss_aed`: Computed loss
- `state`: green / amber / red

**2. Daily Reading (`reading`)**

- `site_id`: Target site link
- `date`: Reading date
- `sun_hours`: Actual sunlight hours
- `actual_kwh`: Inverter production
- `expected_kwh`: Cap × Sun × PR
- `lost_kwh`: Expected - Actual
- `lost_aed`: lost_kwh × Tariff
- `is_post_clean`: Recovery flag

**3. Cleaning Job (`dustwatch.job`)**

- `site_id`: Target rooftop link
- `trigger_loss_aed`: Loss at trigger
- `state`: proposed / scheduled / done
- `date_scheduled`: Planned date
- `date_done`: Completion date
- `recovery_kwh_day`: Output restored
- `payback_days`: Cost / Daily recovery
- `invoice_id`: Optional invoice link

### 48-Hour Hackathon Delivery Scope Matrix

| Component | Must-Have (Core MVP for Demo) | Stretch Goals (If Ahead of Schedule) |
|---|---|---|
| **Data Models** | Site, Daily Reading, Cleaning Job models in Odoo 19 | Customer self-service portal view |
| **Calculation Engine** | Formulas 6.1–6.5 (Expected, Loss in kWh & AED, Trigger) | Formula 6.7 (Dynamic PR recalibration) |
| **UI & Dashboards** | Color-coded Site Tree, 2-line line chart, Kanban board | One-click Customer Invoice generation |
| **Demo Scenario** | 90 days of simulated readings for 3 sites + sandstorm spike | Hardware fault detection flag |

## 7. PITCH DECK PLAYBOOK (FOR THE CANVA POD)

Judges score hackathons on problem severity, technical execution, and commercial viability. This 12-slide master storyboard guides the Canva team to construct a winning 5-minute presentation:

| Slide | Slide Title & Theme | Canva Layout & Visual Assets | Presenter Key Soundbite |
|---|---|---|---|
| 01 | **Title & Hook** | Dramatic split photo: gleaming desert solar farm vs dust-choked panels. | *"Desert solar has an invisible thief: airborne sand."* |
| 02 | **The Regional Crisis** | Bold statistics: 12.7% drop in 5 mos; 27% loss in March; 725 MW Shams Dubai. | *"In the UAE, dust accumulation is not a nuisance—it's an economic crisis."* |
| 03 | **The Maintenance Trap** | Balancing scale: Cleaning too rarely (lost kWh) vs too often (wasted cash & water). | *"Clean too rarely, you lose power. Clean too often, you burn cash and water."* |
| 04 | **Enter DustWatch** | DustWatch logo, sleek UI mockups on laptop and iPad, Odoo badge. | *"Today owners clean by the calendar. DustWatch lets them clean by the numbers."* |
| 05 | **The 6-Step Loop** | Circular process diagram: Reading → Expected → Loss → Trigger → Clean → Verify. | *"An autonomous loop that converts sensor telemetry into scheduled field maintenance."* |
| 06 | **The Golden Math** | U-shaped total cost curve showing global minimum at Cumulative Loss = Cleaning Cost. | *"We mathematically trigger cleaning at the exact financial breakeven point."* |
| 07 | **Live Demo Hand-off** | Full-bleed graphic slide: "Live Odoo Demonstration: Dubai Rooftop Case Study". | *"Let's see this live inside Odoo right now with a real Dubai hotel rooftop."* |
| 08 | **Economic Case Study** | Comparison table: Weekly (AED 80k) vs Quarterly (AED 35k) vs DustWatch (AED 26.5k). | *"DustWatch saves AED 8,500/year on a single roof—a 24% cost cut."* |
| 09 | **Portfolio Scalability** | Infographic showing a 50-roof solar maintenance contractor saving AED 425,000 annually. | *"For solar service companies, this is an instant retention and margin driver."* |
| 10 | **Water & ESG Impact** | Water droplet icons: 43 washes saved = thousands of liters of desalinated water preserved. | *"Fewer unnecessary washes means preserving precious UAE water resources."* |
| 11 | **Odoo Ecosystem Fit** | Process map: Odoo CRM/Sales → Installation → **DustWatch (20-Year Upkeep)**. | *"Odoo sells and installs solar. DustWatch manages the next 20 years of operations."* |
| 12 | **Conclusion & Q&A** | Team credentials, GitHub repository QR code, bold final summary motto. | *"Maximizing clean energy yield, one rooftop at a time. Thank you!"* |

### Canva Design Language & Visual Guidelines

**Color Palette**
**Deep Green:** `#064e3b` (Sustainability)
**Solar Amber:** `#f59e0b` (Sunlight / Alerts)
**Slate Black:** `#0f172a` (Tech Contrast)
**Desert Sand:** `#d97706` (Regional Context)

**Typography & Tone**
**Headers:** Bold Sans (Montserrat / Plus Jakarta).
**Body:** High legibility (Inter / Roboto).
**Tone:** Commercially grounded, engineering-backed, and sustainability-focused.

**Key Visual Metaphors**
**1. Calendar vs Calculator:** Guesswork vs math.
**2. The Inverted Curves:** Visual breakeven.
**3. Water Drops & Coins:** Dual resources saved.

## 8. FINANCIAL ROI & ENVIRONMENTAL WATER SAVINGS

Annual commercial comparison for a standard 500 kWp commercial rooftop installation in Dubai:

| Cleaning Strategy | Annual Cleans | Cleaning Invoices | Lost Solar Revenue | Total Cost of Dust | Economic Outcome |
|---|---|---|---|---|---|
| **Weekly Cleaning** (Too Often) | 52 cleans | AED 78,200 | AED 2,200 | AED 80,400 | **+ AED 53,900 overspent** |
| **Quarterly Cleaning** (Too Rarely) | 4 cleans | AED 6,100 | AED 28,900 | AED 35,000 | **+ AED 8,500 wasted** |
| **DustWatch Optimizer (Dynamic ~6 wks)** | **~9 cleans** | **AED 13,300** | **AED 13,300** | **AED 26,500** | **AED 8,500 Net Savings (24% Cut)** |

### Dual Commercial Stakeholders

**Building Owners:** Saves AED 8,500/year per roof with zero upfront hardware costs.

**O&M Contractors:** Managing 50 rooftops delivers **AED 425,000/year** in verified customer savings, driving retention and contract expansions.

### Environmental & ESG Benefits

**Water Preservation:** Eliminating ~43 redundant washes per roof saves tens of thousands of liters of scarce desalinated water.

**Grid Decarbonization:** Recovers ~15,000 kWh of lost solar energy, displacing fossil fuel peaker plant generation.

## 9. LIVE DEMO WALKTHROUGH & JUDGE Q&A DEFENSE

The presentation seamlessly transitions into the 3–minute live Odoo demonstration:

1. **Portfolio Overview:** Open Odoo Site List. Point out the red badge on "Grand Hyatt Rooftop" (AED 1,515 loss).
2. **Root-Cause Chart:** Click site. Show the 2–line chart: output sagging over 40 days, then plunging post-sandstorm.
3. **Autonomous Dispatch:** Switch to Cleaning Kanban. Highlight the Job DustWatch automatically generated.
4. **Verification & Reset:** Drag job to "Done", load the next week of readings, show actual output recover to 100%, and point out calculated payback: *"This clean paid for itself in 23 days."*

**Q: "Where do daily readings come from?"**

**Defense:** Standard inverters (SMA, Huawei, Sungrow) already stream daily kWh via Modbus/API. Sun hours come from satellite APIs or on-site pyranometers. We ingest via CSV for the demo to validate the business logic.

**Q: "What happens when it rains in Dubai?"**

**Defense:** Rain washes panels for free. Generation rebounds immediately, daily loss drops to zero, and the running total halts. DustWatch automatically postpones cleaning without requiring manual intervention.

**Q: "How do you distinguish equipment faults from dust?"**

**Defense:** Two safeguards: 1) Dust accumulates gradually (0.2%/day) vs inverter failure which is immediate; 2) Post-clean recovery audit: if output does not rebound after cleaning, DustWatch flags a hardware fault inspection.

**Q: "Why build inside Odoo vs a standalone SaaS?"**

**Defense:** Solar installers already run on Odoo for CRM, Field Service, and Invoicing. A standalone script is just another dashboard nobody checks. DustWatch turns insights directly into dispatched work orders and customer invoices.

**Academic Citations:** Hachicha et al., "Impact of dust on solar PV under UAE weather", Renewable Energy (12.7%). "Global analysis of optimal cleaning cycle", Applied Energy (27% in March). "Influence of cleaning frequency", MDPI (8–34 day range). DEWA Shams Dubai (725 MW, 8,430 buildings).